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Linear Algebra

2Week [11 September - 17 September]

Quiz 2

Started on	2023-09-17 23:51
State	Finished
Completed on	2023-09-18 00:05
Time taken	13 mins 9 secs
Marks	4.50/5.00
Grade	1.80 out of 2.00 (90%)

Question 1

Correct

Mark 1.00 out of 1.00

Define a matrix of transformation that projects a point from R^3

onto Oz -axis.

Select one or more:

☐ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

☒ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ✓

☐ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Your answer is correct.

The correct answer is:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 2

Correct

Mark 1.00 out of 1.00

Find true statements.

Select one or more:

- ☒ The columns of any 4×5 matrix are linearly dependent. ✓
- ☒ If x and y are linearly independent, and if $\{x, y, z\}$ is linearly dependent, then z is in $\text{Span}\{x, y\}$. ✓
- ☐ If S is a linearly dependent set, then each vector is a linear combination of the vectors in S .
- ☒ The columns of a matrix A are linearly independent if the equation $Ax = 0$ has only trivial solution. ✓

Your answer is correct.

The correct answers are: The columns of a matrix A are linearly independent if the equation $Ax = 0$ has only trivial solution., The columns of any 4×5 matrix are linearly dependent., If x and y are linearly independent, and if $\{x, y, z\}$ is linearly dependent, then z is in $\text{Span}\{x, y\}$.

Question 3

Correct

Mark 1.00 out of 1.00

Define a matrix of transformation that rotates any point in R^2 for 90° with counterclockwise rotation.

Select one:

- ☐ $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
- ☒ $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ ✓
- ☐ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- ☐ $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

Your answer is correct.

The correct answer is: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Question 4

Correct

Mark 1.00 out of 1.00

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$, define the image of x under A .

Select one:

☒ $\begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ ✓

☐ $\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$

☐ $\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$

☐ $\begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$

☐ $\begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}$

Your answer is correct.

The correct answer is:

$$\begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$$

Question 5
Partially correct
Mark 0.50 out
of 1.00

Find true statements.

Select one or more:

- ☒ If $v_1, \dots, v_4$ are in R^4 and $v_3 = 2v_1 + v_2$, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent. ✓
- ☐ If $v_1, \dots, v_4$ are in R^4 and $v_3 = 0$, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.
- ☒ If v_1 and v_2 are in R^4 and v_2 is not a scalar multiple of v_1 , then $\{v_1, v_2\}$ is linearly independent. ✗

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: If $v_1, \dots, v_4$ are in R^4 and $v_3 = 2v_1 + v_2$, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.,
If $v_1, \dots, v_4$ are in R^4 and $v_3 = 0$, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.